

Supporting Data and Code: Reproducibility Documentation

Evaluating LLM-Based Credit Rating Systems: A Pre-Registered Specification-Curve Analysis

Companion to the manuscript submission

1. Purpose and scope

This document describes the supporting data-and-code archive that accompanies the manuscript. The archive is a self-contained, **offline**, **deterministic** reproduction: from a single command it regenerates every reported number, table, and figure from frozen inputs, with **no network access, no API keys, and no paid model calls**. The permanent citable version is the Zenodo archive at 10.5281/zenodo.21953934 (code MIT-licensed); the same tree is mirrored at <https://github.com/samirrc2/llm-credit-rating-specification-curve>.

A strict wall separates two phases. An *offline analysis* reproduces every claim from frozen inputs (documented here). A *live capture* phase originally queried commercial model APIs and is retained for transparency only (Section 8). Because commercial models change over time, capture is deliberately not part of the reproducible core; the frozen model-response corpora it produced are shipped and hash-verified. The confirmatory data are synthetic firm profiles benchmarked to an objective Altman Z'' reference; the real-firm arm uses only publicly available SEC filings, anonymized and perturbed.

2. Quick start

```
pipinstall-rrequirements.txt
bashreproduce.sh
```

Expected result: the full analysis regenerates every `results/results_*.json`; a second independent run is byte-identical (the SHA-256 of each JSON result matches), and the run exits 0. Total runtime is a few minutes on a standard laptop. No network access is required at any point.

3. Archive layout

<code>reproduce.sh</code>	one-command offline reproduction of all results
<code>requirements.txt</code>	pinned Python dependencies (analysis only; Python 3.10)
<code>config/</code>	specification grid, paraphrase templates, rating scale, run config
<code>analysis/</code>	all analysis code (Section 5)
<code>capture/</code>	live data-collection harness (reference only; not run here, Section 8)
<code>data/frozen/main/</code>	pre-registered inputs: <code>battery_90.json</code> , <code>grid_definition.json</code> , <code>paraphrase_templates.json</code> , <code>rating_scale.json</code> , <code>capital_map.json</code> , manifests
<code>data/frozen/realarm/</code>	real-firm arm artifacts: anonymized battery, sealed crosswalk, rating provenance, 12-spec grid

data/frozen/market_data/	FRED option-adjusted-spread series (cached)
data/raw/main/	frozen response corpus (per-call JSON)
data/raw/realarm/	real-firm arm model-output panels (arm / comparator / fingerprint)
capture/flagship_runs/	flagship model-tier arm corpus (1,620 per-call JSON), fixed by MANIFEST_FLAGSHIP
capture/reasoning_runs/	reasoning-architecture arm corpus (DeepSeek-R1; 1,620 per-call JSON + consolidated panel), fixed by MANIFEST_REASONING
data/panel/panel.parquet	analysis panel, built once from raw (the confirmatory analysis input)
results/	all generated outputs (Section 6)
paper/ISWA/	manuscript (Elsevier CAS), figures, highlights, cover letter, .docx
manifest/	SHA-256 manifests for every frozen stage
docs/	reports, appendix, changelog
PREREGISTRATION.md, PREREGISTRATION_AMENDMENTS.md	frozen pre-registration and amendments

4. Data (inputs and metadata)

4.1 Frozen inputs

Every input the pipeline reads is a frozen artifact. The analysis is a pure, seeded function of these files.

File	Rows	Unit of observation / key fields
data/frozen/main/battery_90.json	90	one synthetic firm profile; 45 credit-health items (SAFE/WATCH/DISTRESS, 15 each) with an Altman Z'' benchmark band + 45 directional items (buy/hold/sell). Fields: <code>item_id</code> , <code>family</code> , <code>benchmark_label</code> , <code>facts_kv</code> , <code>facts_text</code>
data/raw/main/run_20260707_185649/	12,960	one row = one model call across the pre-registered grid; the confirmatory analysis uses the two providers with complete current+prior coverage (OpenAI, Google) = $90 \times 32 \times 3 = 8,640$ cells. An excluded third-provider pilot is retained for transparency. Fields: <code>spec_axes</code> , <code>item_id</code> , <code>seed</code> , <code>provider</code> , <code>model</code> , <code>raw_response</code>
data/frozen/realarm/battery_real45.json	31	anonymized, perturbed real B2B issuers with disclosed agency ratings; benchmarked in <code>ratings_provenance.csv</code> (verbatim 10-K quote, SEC URL, document SHA-256)
data/frozen/realarm/spec_grid_12.json	12	the frozen 12-specification grid (presentation $\times$ provider $\times$ paraphrase $\times$ format) shared by the real-firm and flagship arms

File	Rows	Unit of observation / key fields
capture/flagship_runs/run_20260830_141844/	1,620	flagship model-tier arm: gpt-5.4 + gemini-3.1-pro-preview on the 12-spec grid $\times$ 45 credit items $\times$ 3 seeds
capture/reasoning_runs/run_20260906_111300/	1,620	reasoning-architecture arm: deepseek-r1 on the 12-spec grid $\times$ 45 credit items $\times$ 3 seeds (reasoning_panel.json is the consolidated input read by reasoning_analysis.py)
config/grid_definition.json	—	the balance-constrained D-optimal specification grid
data/frozen/main/rating_scale.json	—	21-grade letter scale and the letter $\rightarrow$ notch/band/IG-HY maps
data/frozen/main/capital_map.json	—	Basel CRE20 ECRA risk-weight map (retained for provenance; the post-hoc capital analysis was removed in the major revision and is no longer part of the pipeline)

4.2 Provenance and integrity

manifest/ holds a SHA-256 manifest for every frozen stage: MANIFEST_INPUTS (pre-registered inputs), MANIFEST_RAW (response corpus), MANIFEST_PANEL (analysis panel), MANIFEST_ANALYSIS, MANIFEST_CAPITAL, MANIFEST_MARKET, MANIFEST_REALARM, and MANIFEST_FLAGSHIP (flagship arm). PREREGISTRATION.md and PREREGISTRATION_AMENDMENTS.md freeze the design, so any silent change to an input or to the pre-registered plan is detectable before the analysis runs. Post-hoc analyses (the flagship tier arm, the self-consistency curve, and the score validation) are labelled as such in both the manuscript and this document.

4.3 Data sources

All confirmatory inputs are synthetic and self-contained: the 90-item battery is frozen and its Altman Z'' benchmark is independently computable before any model is queried. The real-firm arm uses only public SEC 10-K filings and XBRL company facts (keyless EDGAR; permanent URLs in ratings_provenance.csv); the disposable build cache is not shipped and is not needed to reproduce any result. The original capture queried the OpenAI and Google APIs; those responses are frozen under data/raw/ and capture/flagship_runs/ and are the only trace of the live phase the reproduction consumes.

5. Code (what it consumes and runs)

5.1 Analysis scripts

All code is in analysis/; each module is deterministic (seed 42) and reads only frozen data.

Script	Purpose	Produces
run_analysis.py	confirmatory analysis: OLS Type-II ANOVA variance decomposition (item / axes / provider×item / pure-seed / other), within/cross-vendor agreement, granularity κ , IG/HY flip	results.json, spec_curves.json, MANIFEST_ANALYSIS
addenda.py	supplementary robustness figures/numbers	results_addenda.json
benchmark_validation.py	power / benchmark validation checks	benchmark_validation.*
analyze_realarm.py	real-firm arm: fingerprint gate, WATCH-stratum flip-share, descriptive arm-vs-battery comparison (no equivalence test), band accuracy vs agency	MANIFEST_VALIDATION_H1_results.json
realarm_speccurve.py	external-benchmark specification curve on real issuers (agency ratings; issuer-preserving permutation)	results_realarm_speccurve*.json
realarm_scale_sensitivity.py	does instability track absolute dollar scale? (Z'' -invariance, Spearman)	results_scale_sensitivity.json
tier_analysis.py	flagship vs nano/flash IG/HY & band flip on the 12-spec grid, issuer-clustered CIs	results_tier.json
extended_analysis.py	specification-instability score validation (ROC-AUC, held-out & real-issuer), self-consistency curve, ordinal-weighted agreement, per-stratum flips, deployment cost	results_extended.json
revision_analysis.py	boundary-distance severity (R3-7) and turnover recast (R3-6)	results_revision.json
permtest_and_tau.py	rating-invariance permutation null (R3-3) and SIS τ calibration (R3-5)	results_permtest_tau.json
variance_model_range.py	range of variance explained by elicitation axes (main-effects / +2-way / issuer-OOS ridge)	results_variance_model_range.json
crossmodel_aliasing.py	cross-model transfer of the instability score and design aliasing diagnostics (R2-4/R2-5/R3-2)	results_crossmodel_aliasing.json
reasoning_analysis.py	reasoning-architecture arm (DeepSeek-R1) vs nano/flash & flagship on the frozen grid	results_reasoning.json
build_panel.py	(audit) rebuild <code>panel.parquet</code> from the raw corpus	panel (row-identical)
make_figures.py, make_extended_figures.py	regenerate manuscript figures (optional; needs matplotlib)	paper/ISWA/*.pdf,png

5.2 Pipeline order

`reproduce.sh` runs the modules in dependency order; all JSON outputs are timestamp-free, so re-runs are byte-identical:

```
run_analysis→addenda→benchmark_validation→analyze_realarm→realarm_speccurve→realarm_scale_sensitivity→tier_analysis→extended_analysis→revision_analysis→permtest_and_tau→variance_model_range→crossmodel_aliasing→reasoning_analysis
```

6. Results (generated outputs)

Running the pipeline regenerates the following under **results/**. In a clean checkout these already exist; the reproduction overwrites them and confirms they are unchanged.

Output	Contents
<code>results.json</code>	headline confirmatory numbers: variance shares (with issuer-clustered CIs), agreement ladder (within 82.5% / cross 64.4%), IG/HY matched-pair disagreement (25.2%)
<code>spec_curves.json</code>	per-specification label-recovery / IG-share for the specification-curve figure
<code>results_permtest_tau.json</code>	rating-invariance permutation null (temperature-0-honored subgrid: observed 1.59, $p = 0.0005$, $\eta^2 = 0.46$) and SIS τ calibration ($\tau = 0.08$)
<code>results_tier.json</code>	flagship vs nano/flash flip shares and paired differences (−1.1 pp, 95% CI −7.6 to +6.0)
<code>results_reasoning.json</code>	DeepSeek-R1 reasoning arm: IG/HY flip 25.5% (95% CI 20.9–30.1), paired diff vs nano/flash +0.9 pp
<code>results_extended.json</code>	score validation (held-out ROC-AUC 0.94, CI 0.86–0.99; real-issuer 0.70), self-consistency curve ($k = 1 \dots 15$), Krippendorff α , per-stratum flips, deployment cost
<code>results_crossmodel_aliasing.json</code>	cross-model transfer (Spearman 0.14–0.73, median 0.45; AUC 0.72) and aliasing (main off-diag 0.062; main×2FI 0.333)
<code>results_variance_model_range.json</code>	variance explained by elicitation axes: 0.083 / 0.119 / 0.101 (main / +2-way / issuer-OOS)
<code>results_realarm_speccurve_*.json</code>	external-benchmark spec curve on real issuers (42.2–61.1%, mean 53.7%, perm $p = 0.021$)
<code>results_scale_sensitivity.json</code>	scale-invariance check (Spearman −0.035, perm $p = 0.85$)
<code>results_revision.json</code> , <code>results_addenda.json</code>	boundary-distance severity, turnover recast, supplementary robustness
<code>H1_results.json</code>	real-firm arm descriptive outputs (WATCH flip-shares, arm–battery difference, band accuracy)
<code>benchmark_validation.{json, csv}</code>	power / benchmark validation
<code>results/exhibits/</code>	regenerated tables

7. Reproduction and verification

`reproduce.sh` performs, and exits non-zero if any stage fails:

1. **Integrity.** Each frozen stage carries a SHA-256 manifest (`manifest/`); the raw-corpus freeze rejects any capture file above the pre-set error threshold (observed non-substantive

rate 0.08%, 7 of 8,640 confirmatory cells, dropped listwise).

2. **Regeneration.** The full analysis (Section 5.2) runs from the frozen inputs only.
3. **Byte-identical re-run.** Re-running `reproduce.sh` regenerates every `results/results_*.json` byte-for-byte (verified); the results of record are the committed JSON files.

Determinism. All stochastic steps use fixed seeds (analysis seed 42; issuer-clustered bootstraps use fixed-seed resamples), so confidence intervals are exactly reproducible. *Note:* Parquet serialization is not byte-deterministic, so a rebuilt `panel.parquet` may hash differently while containing row-identical data; the confirmatory analysis reads the released frozen panel, so every reported number reproduces exactly.

Environment. Pinned analysis environment: `numpy==2.2.6`, `pandas==2.2.3`, `pyarrow==24.0.0`, `statsmodels==0.14.6`, `scipy==1.15.3`, `patsy==1.0.2`, on Python 3.10 (exact pins in `requirements.txt`). Figures are optional and need `matplotlib`, which is not part of the pinned analysis dependencies.

8. Live data collection (not part of the reproduction)

`capture/` documents how the model-response corpora were originally collected: `collect.py` drives the main confirmatory grid; `run_arm.py` and `realarm_collect.py` run the real-firm arm; `run_flagship.py` runs the flagship model-tier arm and `freeze_flagship.py` writes its SHA-256 manifest; `run_reasoning.py` runs the reasoning-architecture arm (DeepSeek-R1 via OpenRouter) and `freeze_reasoning.py` writes `MANIFEST_REASONING`; `fingerprint_pilot.py` runs the perturbation-and-fingerprint-screening gate; `list_models.py` enumerates served models. This phase requires internet access and provider API keys, incurs cost, and is intentionally excluded from the reproducible core. API keys are never distributed (`.gitignore` blocks `keys.env.txt` and any `API Keys/` folder); all reported results derive only from the frozen corpora.

9. Licensing and data availability

Code is released under the MIT Licence (see `LICENSE`); data, analysis outputs, and text are released under CC-BY-4.0. The confirmatory data are synthetic; the real-firm arm uses only public SEC filings (anonymized and perturbed). The permanent citable archive is the Zenodo deposit at `10.5281/zenodo.21953934`, which reproduces every reported number, table, and figure offline, deterministically, and at zero cost.